

Education

Expected Jun 2027 **B.S. in Statistics, Data Science Track**, *University of Washington*, Seattle, WA.

Expected Jun 2027 **B.S. in Informatics, with Honors**, *University of Washington*, Seattle, WA.

Cumulative GPA: 3.9/4.0

Selected coursework: Statistical Machine Learning, Data Programming, Data Structures and Algorithms, Statistical Computing, Databases and Data Modeling, Mathematical Statistics & Probability I-III, Generalized Linear Models, Real Analysis.

Research Experience

University of Illinois Urbana-Champaign

Summer 2026 **Research Intern, Data Mining Group**.

Lead research on AI for science, focusing on retrieval-augmented generation for retrosynthesis and drug discovery. Build and evaluate LLM/retrieval workflows for chemical reasoning and scientific decision support.

Advisor : **Prof. Jiawei Han**, *Michael Aiken Chair Professor, Siebel School of Computing and Data Science, UIUC* ([Personal Web-page](#))

University of Washington

Nov 2025 – Present **Research Assistant, InfoSeeking Lab**.

Conduct research on epistemic AI agents for information seeking and human-AI collaboration. Design benchmarks and model-based evaluations for agents that model users' structural ignorance, navigate knowledge gaps, and build stronger epistemic partnerships under uncertainty.

Advisor : **Prof. Chirag Shah**, *Professor, Information School; Adjunct Professor, Paul G. Allen School of CSE, University of Washington* ([Personal Web-page](#))

Apr 2026 – Present **Research Collaborator, UW NLP Group**.

Co-lead research on benchmarking whether LLM assistants understand users' well-being, preferences, and conflicts between short-term requests and long-term goals. Develop evaluation protocols and training-oriented analyses for proactive, personalized assistant behavior.

Collaborator : **Stella Li**, *Ph.D. Student, Paul G. Allen School of CSE, University of Washington* ([Personal Web-page](#))

Publications & Workshops

Manuscripts

Manuscript Kaur, Kirandeep; **Lyu, Xingda**; Zhou, Zhiyu; Kleiman-Weiner, Max; and Shah, Chirag, 2026. *Modeling Evolving Minds: Toward Epistemic Human-AI Collaboration*. Manuscript in preparation.

Manuscript **Lyu, Xingda**; Chan, Joey; and Han, Jiawei, 2026. *RetroRAG: Multi-Layer Retrieval-Augmented Generation for Retrosynthesis Prediction*. Manuscript in preparation.

In Conference Proceedings

ICML 2026 Kaur, Kirandeep; **Lyu, Xingda**; and Shah, Chirag, 2026. *Position: Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight*. In Proceedings of the International Conference on Machine Learning (ICML 2026). ([paper](#))

CDS 2025 **Lyu, Xingda**; Lyu, Gongfu; Yan, Zitai; and Jiang, Yuxin, 2025. *P-RAG: Prompt-Enhanced Parametric RAG with LoRA and Selective CoT for Biomedical and Multi-Hop QA*. In Proceedings of the International Conference on Computing and Data Science (CONF-CDS 2025). ([paper](#))

Workshops

- CVPR 2026 **Lyu, Xingda**; Luo, Xinye; Lu, Honglin; and Yang, Shiqi, 2026. *PROS: A Proactive Multimodal Refinement Agent for Scientific Posters*. Workshop paper, Human-Interactive Generation and Editing (HiGen), CVPR 2026.
- CHIIR 2026 **Lyu, Xingda**, 2026. *Decision-Making Under Unknown Unknowns in Proactive Agentic Systems*. Lightning talk and poster presentation, ACM CHIIR Workshop on Human-Centered Proactive & Personalized Agents.
- ARD 2026 Kaur, Kirandeep; Gupta, Vinayak; **Lyu, Xingda**; Gupta, Aditya; and Shah, Chirag, 2026. *Proactive Agents for Knowledge Gap Navigation*. Poster presentation, Amazon Research Day 2026, AI Co-Scientist. (*poster*)

Professional Experience

- Jun 2025 – Aug 2025 **Data Science Intern**, Amazon, Seattle, WA.
Built automated data-quality pipelines over 300K+ transportation records using Python and SQL. Developed routing optimization and anomaly-detection analyses to identify operational inefficiencies, refund gaps, and underperforming carriers.
- Nov 2024 – May 2025 **Research Project Lead, P-RAG**, Carnegie Mellon University.
Led a research project on retrieval-augmented biomedical and multi-hop question answering. Integrated retrieval, LoRA-based parametric adaptation, selective chain-of-thought prompting, and data augmentation into an experimental QA pipeline. Advisor: **Prof. David P. Woodruff**.
- Jul 2024 – Sep 2024 **Data Science Intern**, Baynovation, San Jose, CA.
Built a refinance decision tool using real-time mortgage rate data. Developed an interactive mortgage-scenario dashboard and delinquency-risk models using GLM, Random Forest, and XGBoost.

Honors & Patents

- Award **iSchool Conference Travel Award**, University of Washington (\$1,500), 2026.
- Patent **CNIPA-granted utility patents** for automated veterinary pharmaceutical production systems, co-inventor, 2023.
- Patent **Chinese invention patent** for a microbial agent for preventing wheat powdery mildew, co-inventor, 2022.

Skills

- Languages Python, R, SQL, Java, JavaScript, HTML/CSS, LaTeX
- ML / AI PyTorch, Hugging Face Transformers, scikit-learn, LoRA/PEFT, RAG, LLM fine-tuning, multi-modal LLM/VLM systems
- Research Information retrieval, human-AI interaction, proactive agents, AI for science, benchmark design, prompt and example design
- Evaluation LLM evaluation, ablation studies, statistical modeling, qualitative error analysis, experimental analysis
- Tools Git/GitHub, Linux, Jupyter Notebook, Flask, Overleaf, PowerPoint/PPTX